GREEEN tallgrass prairie restoration: RLQ analyses pool size (NW Station)

MB

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
```

```r
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33")#, ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "NW Station",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(
    treatment = str_c(seeding_time, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
sites %>%
  group_by(year) %>%
  count()
```

Plots per year

```
## # A tibble: 4 x 2
## # Groups:   year [4]
##   year      n
```

```
##   <fct> <int>
## 1 2015     48
## 2 2016     48
## 3 2017     48
## 4 2018     48
```

```
species %>%
  group_by(id_plot_year) %>%
  count(name = "seeded_forb_richness") %>%
  ungroup() %>%
  count(seeded_forb_richness, name = "number_of_plots") %>%
  mutate(
    "plots_>=_1_species" = sum(number_of_plots),
    total_number_plots = 192
    )
```

Seeded species richness

```
## # A tibble: 7 x 4
##   seeded_forb_richness number_of_plots `plots_>=_1_species` total_number_plots
##                  <int>           <int>                <int>               <dbl>
## 1                    1              65                  143                 192
## 2                    2              43                  143                 192
## 3                    3              18                  143                 192
## 4                    4              10                  143                 192
## 5                    5               4                  143                 192
## 6                    6               1                  143                 192
## 7                    7               2                  143                 192
```

```
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 13 x 4
##    accepted_name                   sla height seedmass
##    <chr>                         <dbl>  <dbl>    <dbl>
##  1 Baptisia alba                 268.   0.940    -4.12
##  2 Eryngium yuccifolium           91.7  1.01     -5.57
##  3 Helianthus occidentalis       184.   0.537    -6.11
##  4 Lespedeza virginica           126.   0.708    -6.26
##  5 Liatris cylindracea           110.   0.564    -5.73
##  6 Lupinus perennis              167.   0.326    -3.84
##  7 Silphium terebinthinaceum     171.   0.716    -4.53
##  8 Solidago juncea               130.   0.783    -7.79
##  9 Solidago nemoralis            122.   0.61     -7.84
## 10 Solidago speciosa             263.   0.71     -8.27
## 11 Symphyotrichum oolentangiense 143.   0.362    -8.68
## 12 Tradescantia ohiensis         103.   0.921    -5.61
## 13 Zizia aptera                  100.   0.34     -6.42
```

Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year") %>%
  select(-water_cap)

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

Descriptive analyses

```
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 4 x 2
## # Groups:   year [4]
##   year      n
##   <fct> <int>
## 1 2015     35
## 2 2016     31
## 3 2017     38
## 4 2018     39
```

Ordinations

```
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```
m_rlq <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

Model 2: species-environment relationship

Model 4: species-traits relationship

```
randtest(m_rlq, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m_rlq, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test       Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.3198689  9.933899 greater  0.001
## 2 Model 4 0.3198689 -1.275169 greater  0.911
```

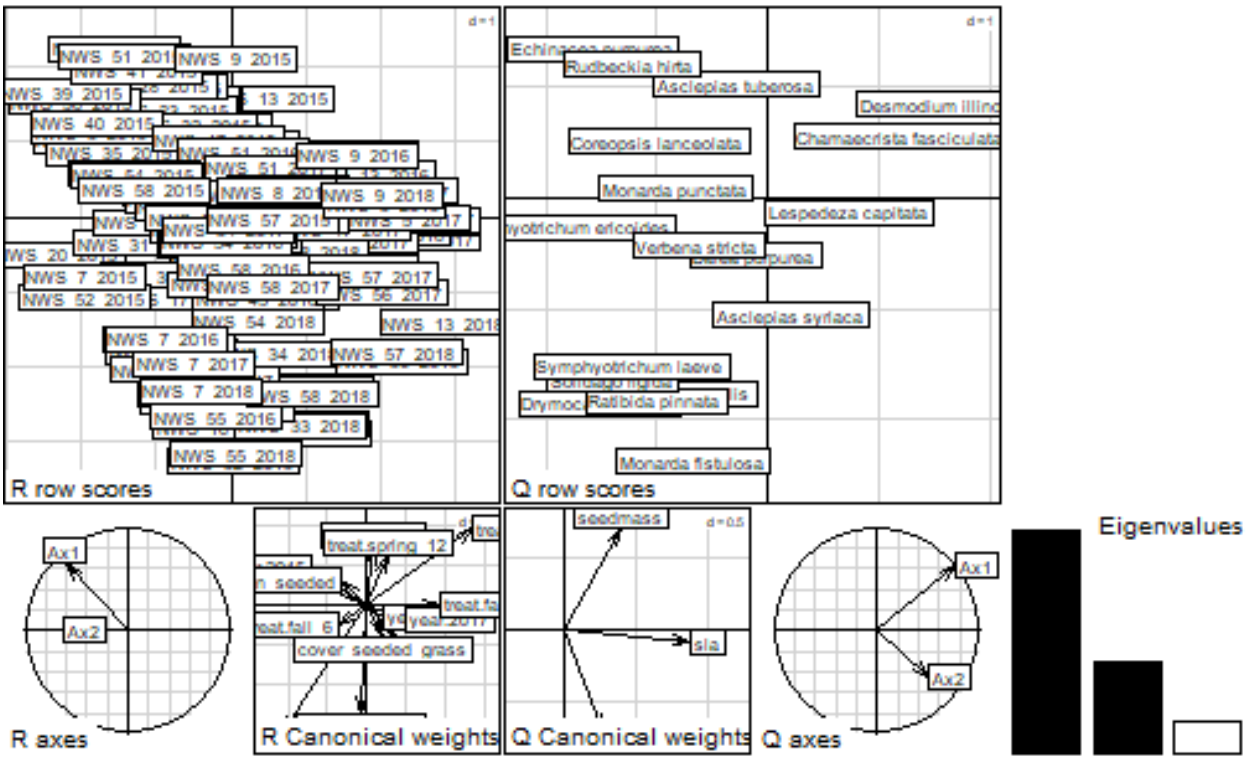
Traits by environment crossed table

```
m_rlq$tab
```

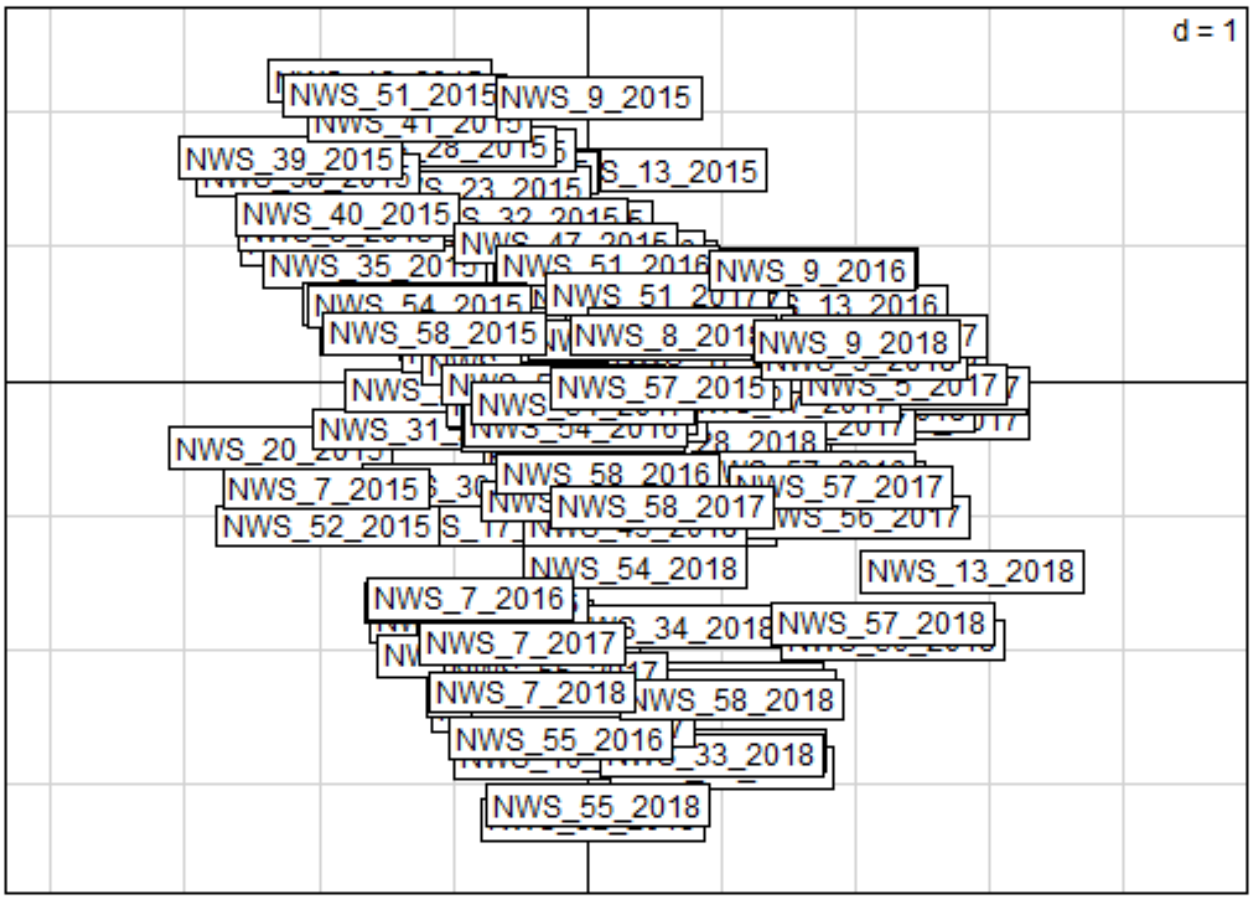
```
##                           sla      height     seedmass
## year.2015          -0.27286366 -0.21226279 -0.057051582
## year.2016           0.12379230  0.01369609 -0.030681192
## year.2017           0.16296480  0.14110756  0.090633359
## year.2018           0.13706422  0.15934892  0.013637862
## treat.fall_12      -0.36933667 -0.09119482 -0.011466502
## treat.fall_18      -0.07048194 -0.02371272  0.261947416
## treat.fall_33       0.40408371  0.06888350  0.121685011
## treat.fall_6       -0.08496939 -0.06932142 -0.189270773
## treat.spring_12     0.13586896 -0.15515897  0.113780798
## treat.spring_18    -0.27254861  0.64886792  0.059466012
## treat.spring_33     0.57228181 -0.11585530  0.355469515
## treat.spring_6     -0.32934008  0.12101077 -0.591223648
## cover_non_seeded   -0.13517428 -0.09727297  0.007148381
## cover_seeded_grass  0.08642838  0.11213277 -0.040272261
```

RLQ Plots

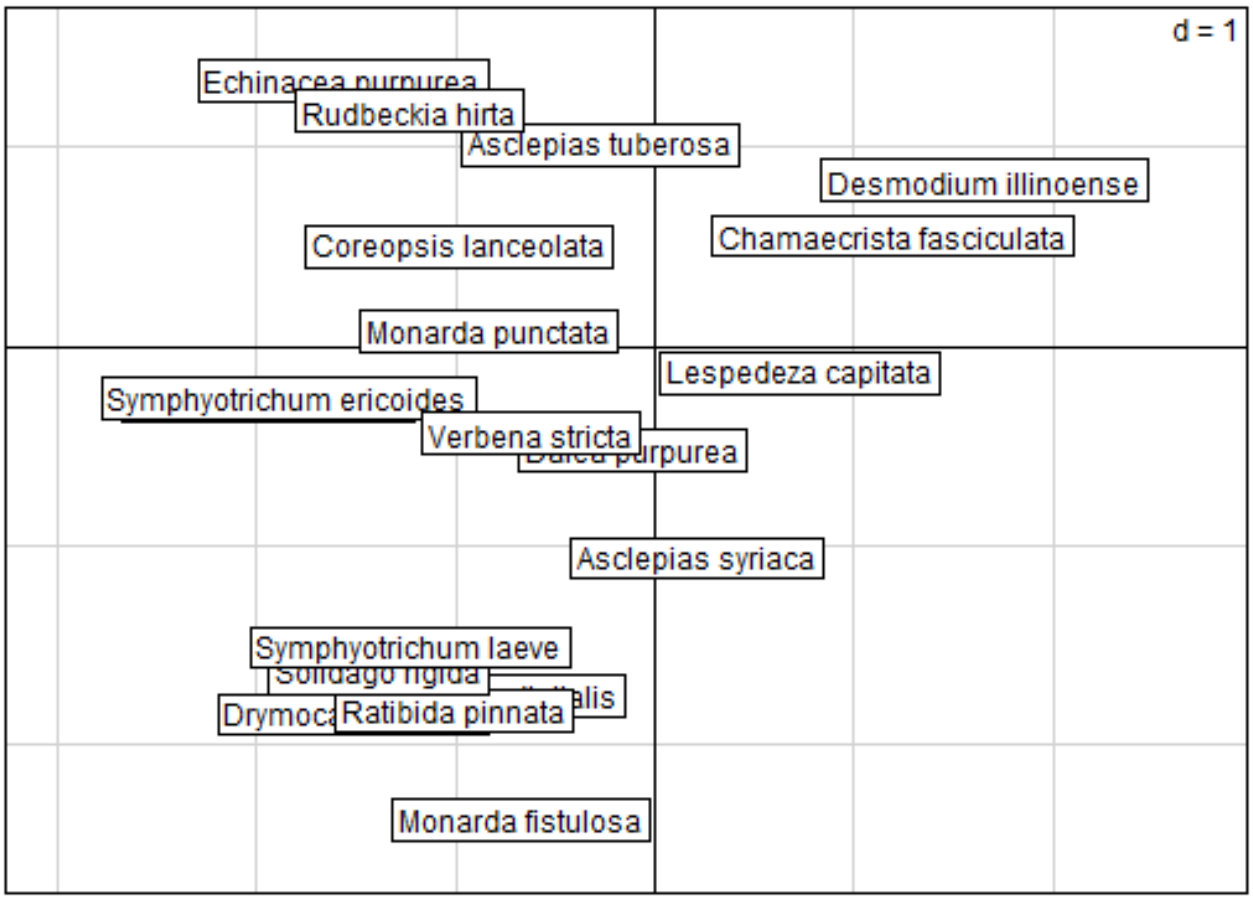
```
plot(m_rlq)
```



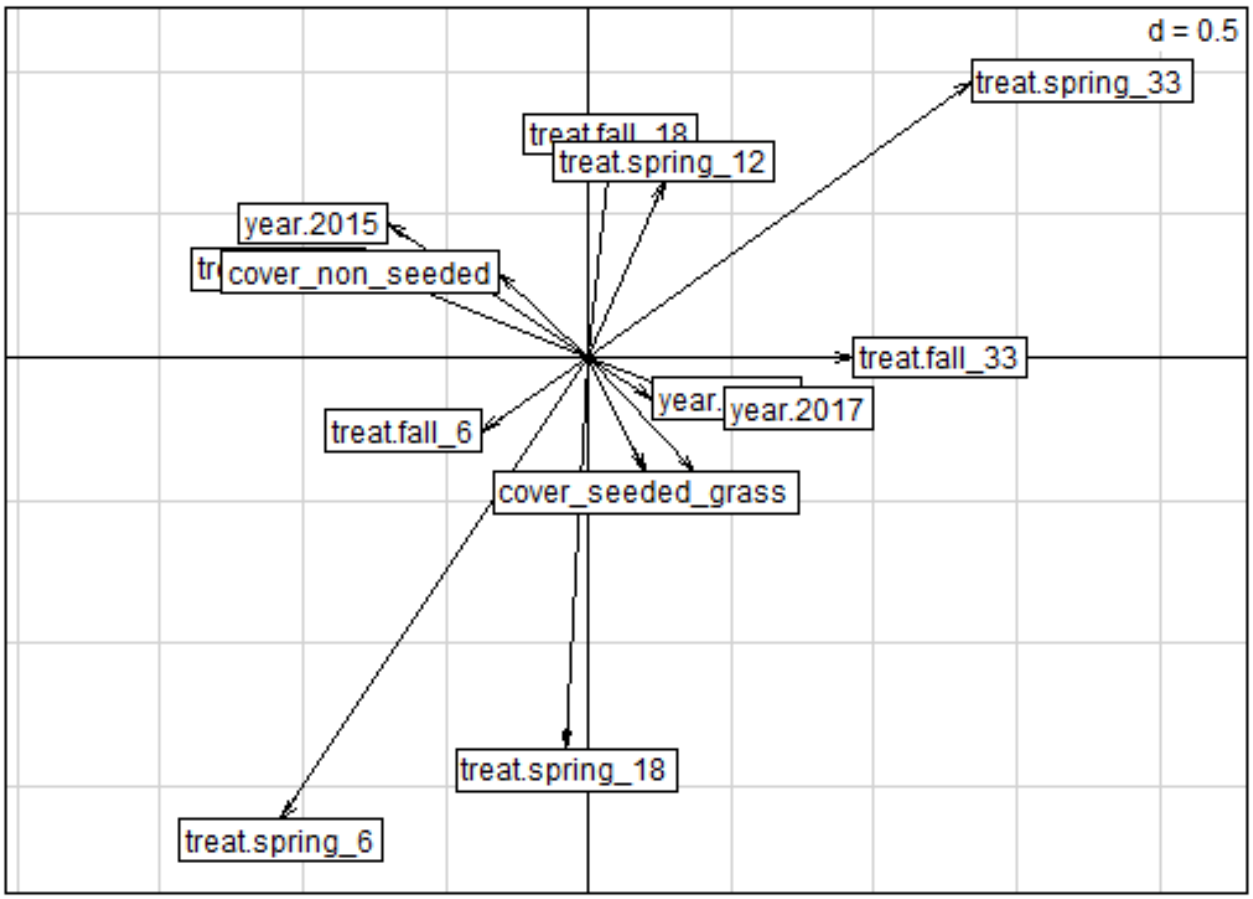

```
s.label(m_rlq$lR)
```



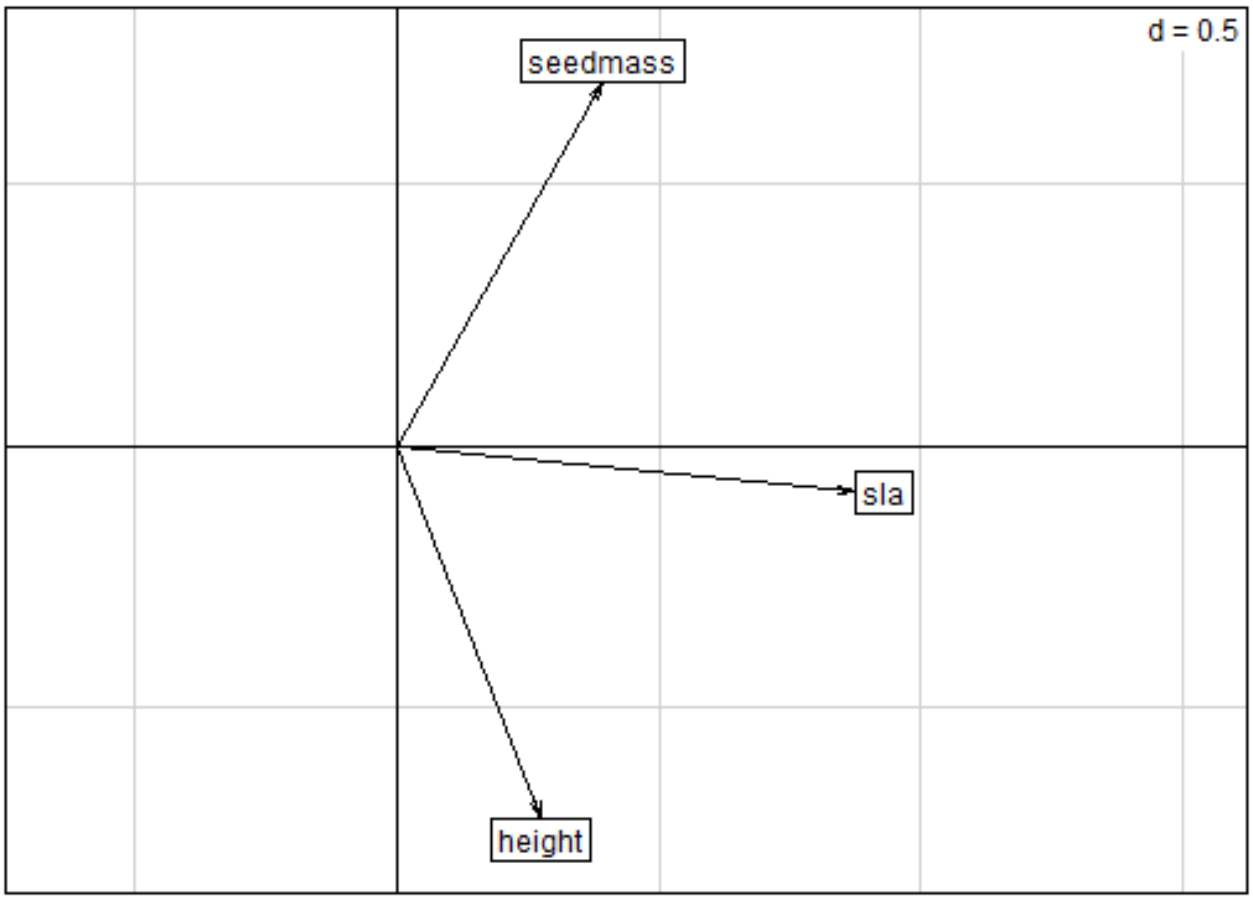

```
s.label(m_rlq$lQ)
```




```
s.arrow(m_rlq$l1)
```




```
s.arrow(m_rlq$c1)
```



Fourth-corner analysis

```
m_4th <- fourthcorner(
  tabR = sites,
  tabL = species,
  tabQ = traits,
  modeltype = 6,
  p.adjust.method.G = "none",
  p.adjust.method.D = "none",
  nrepet = 49999
)
```

```
m_4th_adj <- p.adjust.4thcorner(
  m_4th,
  p.adjust.method.G = "fdr",
  p.adjust.method.D = "fdr",
  p.adjust.D = "global"
)
m_4th_adj
```

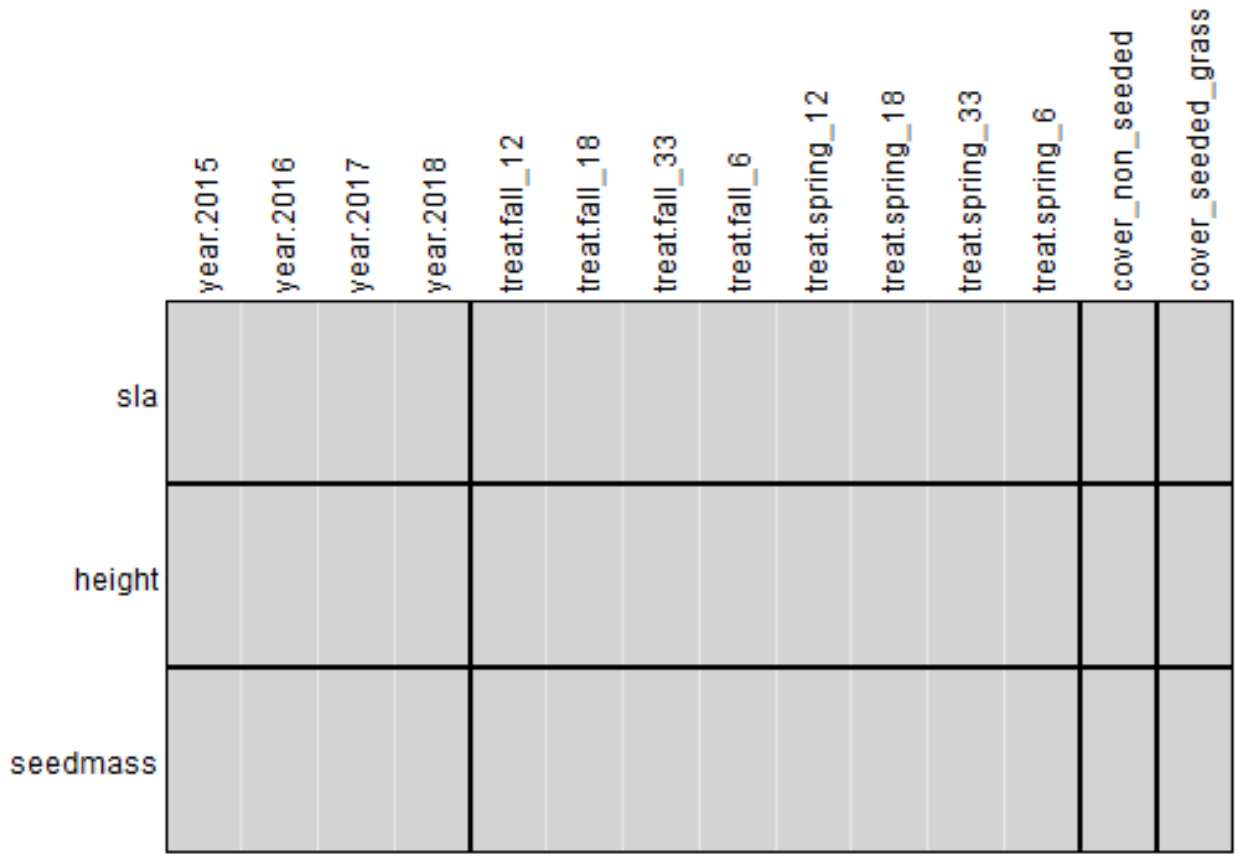
```
## Fourth-corner Statistics
## ------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   fdr
```

```
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs    Std.Obs     Alter
## 1                year.2015 / sla Homog.  0.290604717  1.5755414      less
## 2                year.2016 / sla Homog.  0.188819497  0.4487917      less
## 3                year.2017 / sla Homog.  0.263250987  0.5343451      less
## 4                year.2018 / sla Homog.  0.218131402  0.3023133      less
## 5            treat.fall_12 / sla Homog.  0.071279712 -1.2413297      less
## 6            treat.fall_18 / sla Homog.  0.119767267 -0.4485235      less
## 7            treat.fall_33 / sla Homog.  0.164978031  0.8251609      less
## 8             treat.fall_6 / sla Homog.  0.138829928  0.2651958      less
## 9          treat.spring_12 / sla Homog.  0.068073813  0.6561830      less
## 10         treat.spring_18 / sla Homog.  0.035341663  0.1120562      less
## 11         treat.spring_33 / sla Homog.  0.182050795  3.2318455      less
## 12          treat.spring_6 / sla Homog.  0.118921166  0.2712410      less
## 13        cover_non_seeded / sla      r -0.135174280 -1.3567077 two-sided
## 14      cover_seeded_grass / sla      r  0.086428376  0.9833547 two-sided
## 15            year.2015 / height Homog.  0.599404876  7.1461938      less
## 16            year.2016 / height Homog.  0.139738674 -1.5836013      less
## 17            year.2017 / height Homog.  0.137119436 -2.4743558      less
## 18            year.2018 / height Homog.  0.097616778 -3.4214408      less
## 19        treat.fall_12 / height Homog.  0.200129426  1.1961912      less
## 20        treat.fall_18 / height Homog.  0.137573460  0.1708311      less
## 21        treat.fall_33 / height Homog.  0.154200143  0.2421028      less
## 22         treat.fall_6 / height Homog.  0.155597382  0.5470656      less
## 23      treat.spring_12 / height Homog.  0.081554842  1.5374172      less
## 24      treat.spring_18 / height Homog.  0.033986020  0.2104169      less
## 25      treat.spring_33 / height Homog.  0.084983652  0.5046528      less
## 26       treat.spring_6 / height Homog.  0.129348230  0.5636501      less
## 27     cover_non_seeded / height      r -0.097272965 -0.9443960 two-sided
## 28   cover_seeded_grass / height      r  0.112132768  1.2255894 two-sided
## 29          year.2015 / seedmass Homog.  0.228172198 -0.4886434      less
## 30          year.2016 / seedmass Homog.  0.214705627  1.7113850      less
## 31          year.2017 / seedmass Homog.  0.274660194  1.0952950      less
## 32          year.2018 / seedmass Homog.  0.279108363  2.0881150      less
## 33      treat.fall_12 / seedmass Homog.  0.077864155 -1.6431333      less
## 34      treat.fall_18 / seedmass Homog.  0.115925184 -0.6121794      less
## 35      treat.fall_33 / seedmass Homog.  0.108795302 -0.9942134      less
## 36       treat.fall_6 / seedmass Homog.  0.264713179  4.0067264      less
## 37    treat.spring_12 / seedmass Homog.  0.076791817  1.2348308      less
## 38    treat.spring_18 / seedmass Homog.  0.033762483  0.1752859      less
## 39    treat.spring_33 / seedmass Homog.  0.054002510 -0.3670101      less
## 40     treat.spring_6 / seedmass Homog.  0.189778859  2.7663452      less
## 41   cover_non_seeded / seedmass      r  0.007148381  0.0728080 two-sided
## 42 cover_seeded_grass / seedmass      r -0.040272261 -0.4409907 two-sided
##     Pvalue Pvalue.adj
## 1   0.9333          1
## 2   0.6526          1
## 3  0.69138          1
## 4  0.56612          1
## 5  0.08926   0.749784
## 6  0.35132          1
```

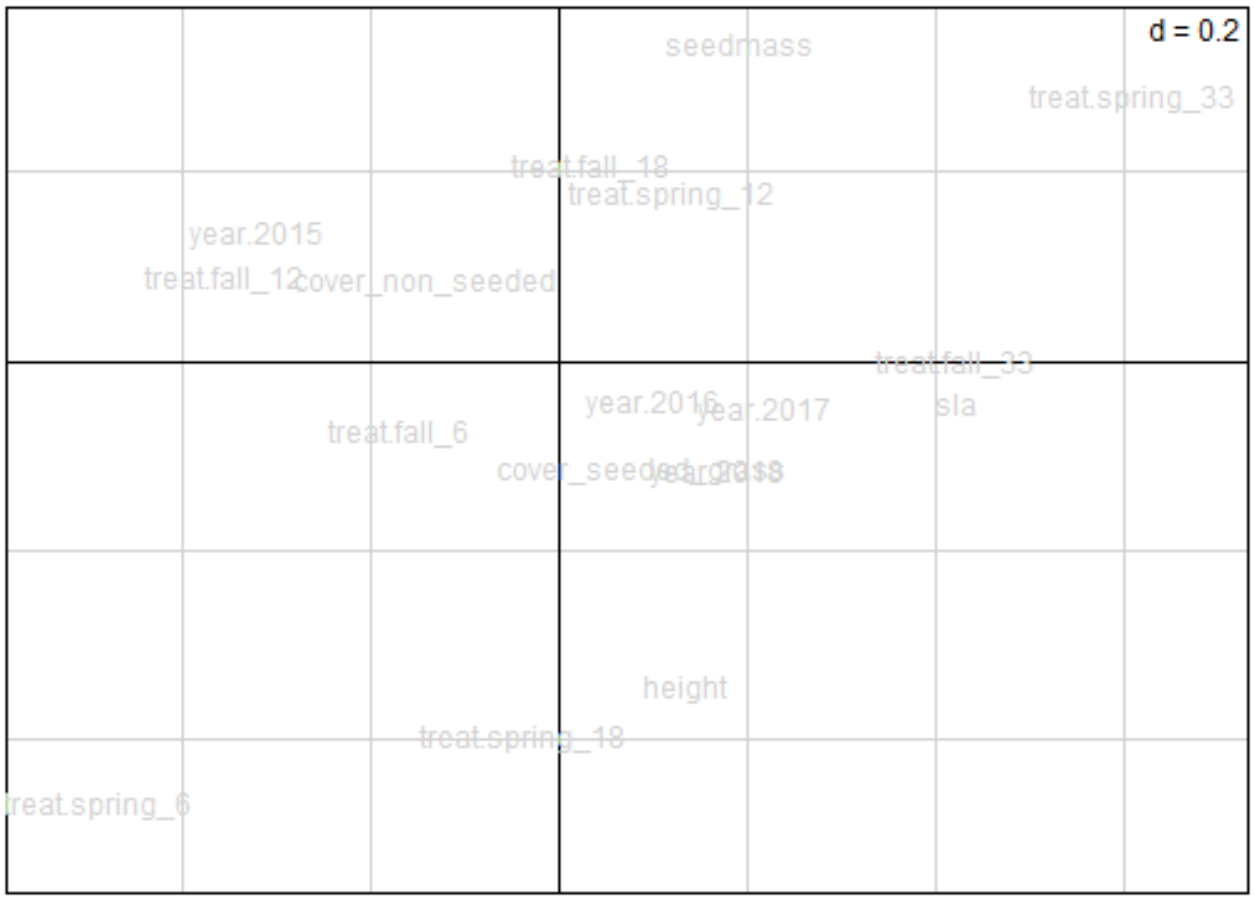
```
## 7  0.79888          1
## 8  0.60598          1
## 9  0.71386          1
## 10 0.59644          1
## 11 0.99724          1
## 12 0.62568          1
## 13  0.1882          1
## 14 0.36036          1
## 15       1          1
## 16  0.0505    0.53025
## 17 0.00394    0.08274 .
## 18 0.00126    0.05292 .
## 19 0.87568          1
## 20  0.5887          1
## 21  0.6123          1
## 22 0.70016          1
## 23 0.93414          1
## 24 0.60996          1
## 25  0.7468          1
## 26 0.70764          1
## 27 0.36618          1
## 28  0.2395          1
## 29 0.30948          1
## 30 0.97606          1
## 31 0.86832          1
## 32 0.99678          1
## 33 0.02886    0.40404
## 34 0.26878          1
## 35 0.15118          1
## 36       1          1
## 37 0.87952          1
## 38  0.6017          1
## 39 0.42872          1
## 40 0.99716          1
## 41 0.94506          1
## 42 0.68192          1
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plot

```r
plot(m_4th_adj, alpha = 0.05, stat = "D2")
```




```
plot(m_4th_adj,
  x.rlq = m_rlq,
  alpha = 0.05,
  stat = "D2",
  type = "biplot"
)
```



Save

Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.1.1    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.1         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```